

BASIC SCIENCE

Identification of oral precancerous and cancerous tissue by swept source optical coherence tomography

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Abstract

Background and Objectives: Distinguishing cancer from precancerous lesions is critical and challenging in oral medicine. As a noninvasive method, optical coherence tomography (OCT) has the advantages of real-time, *in vivo*, and large-depth imaging. Texture information hidden in OCT images can provide an important auxiliary effect for improving diagnostic accuracy. The aim of this study is to explore a reliable and accurate OCT-based method for the screening and diagnosis of human oral diseases, especially oral cancer.

Materials and Methods: Fresh *ex vivo* oral tissues including normal mucosa, leukoplakia with epithelial hyperplasia (LEH), and oral squamous cell carcinoma (OSCC) were imaged intraoperatively by a homemade OCT system, and 58 texture features were extracted to create computational models of these tissues. A principal component analysis algorithm was employed to optimize the combination of texture feature vectors. The identification based on artificial neural network (ANN) was proposed and the sensitivity/specificity was calculated statistically to evaluate the classification performance.

Results: A total of 71 sites of three types of oral tissues were measured, and 5176 OCT images of three types of oral tissues were used in this study. The superior classification result based on ANN was obtained with an average accuracy of 98.17%. The sensitivity and specificity of normal mucosa, LEH, and OSCC are 98.17% / 98.38%, 93.81% / 98.54%, and 98.11% / 99.04%, respectively.

Conclusion: It is demonstrated from the high accuracies, sensitivities, and specificities that texture-based analysis can be used to identify oral precancerous and cancerous tissue in OCT images, and it has the potential to help surgeons in diseases screening and diagnosis effectively.

KEYWORDS

optical coherence tomography, oral cancer, precancerous tissue, texture feature, machine learning, identification

INTRODUCTION

Oral squamous cell carcinoma (OSCC) is one of the most aggressive head and neck cancers, with a local recurrence rate of 25%–50%.¹ Unfortunately, OSCC does not have a desirable prognosis to date and the survival rate of patients has not been improved in decades.² Although the therapeutic methods have been improved in decades, especially surgical methods and chemoradiotherapy, the therapeutic effect is still not satisfactory. Moreover, patients who survive the initial occurrence of oral cancer have an increased risk of developing a second primary

tumor.³ In addition, most of OSCCs arise from potentially malignant lesions, the detection of cancer in this stage can improve the survival rate and prognosis.^{4,5} With early detection, the 5-year survival rate can increase to 83%, while it drops to less than 30% with a delayed diagnosis and the occurrence of metastasis.⁶ Therefore, accurately detecting and identifying oral lesions early, especially distinguishing precancerous and cancerous tissues, remains one of the most urgent issues for improving the oral clinical situation.

Visual inspection and palpation of tissue surface are the conventional screening method for oral lesions.

Generally, the criteria for suspicion of a precancerous or cancerous lesion include changes in surface texture, loss of surface integrity, color, size, contour deviations, or mobility of intra-oral or extra-oral structures.⁷ Besides, a biopsy is performed for highly suspicious tissue regions to provide a definitive diagnosis.⁸ However, the accuracy of the visual examination is highly dependent on the experience of clinicians. Histopathological examination as the gold standard is time-consuming, invasive, and label-intensive.

Most oral lesions originate from the lining of the oral mucosa, such as leukoplakia and OSCC.⁹ Easy access to oral mucosal lesions makes it possible to detect superficial tumors with optical imaging techniques. To assist surgeons in early screening and intraoperative diagnosis, a variety of imaging methods were studied, such as fluorescence imaging,¹⁰ confocal microscopy,¹¹ hyperspectral imaging,¹² and nonlinear microscopy.^{13–15} However, these methods have some limitations and restrict their clinical application. Appropriate fluorescent agents are often required in fluorescence imaging and their safety needs to be carefully verified. Confocal imaging and hyperspectral imaging are usually limited by imaging depth. In contrast, as a noninvasive method, optical coherence tomography (OCT) has the advantages of real-time, *in vivo*, and larger-depth imaging.¹⁶ Since proposed in 1991,¹⁷ OCT has been successfully applied in ophthalmology,¹⁸ cardiology,¹⁹ and gastroenterology.²⁰

In oral medicine, research has been proposed to image oral diseases based on OCT.^{21,22} In addition to structural imaging, there are advancements in OCT image-based recognition methods, which provide objective information to support clinical decision-making. The group of Yang proposed a quantitative OCT-based approach to the diagnosis of oral mucosal tissue.²³ Three indicators were considered, including the standard deviation of A-scans, the exponential decay constant of spatial-frequency spectrum of A-scans, and the epithelium thickness. The results demonstrated that standard deviation and the exponential decay constant are good diagnosis indicators for moderate dysplasia and OSCC, and the epithelium thickness is a good indicator for epithelial hyperplasia and moderate dysplasia.²³ Adegun et al. quantitatively differentiated dysplasia from normal samples by measuring the rate of change of backscattered light intensity with depth in OCT and histological images.²⁴ The previous work of our group showed that OCT images of oral-maxillofacial lesions had obvious distinctions based on different microstructural features.²⁵ On this basis, we investigated the quantitative difference of OSCC and oral non-cancer tissue based on an available optical attenuation model, and the diagnostic sensitivity (97.88%) and specificity (83.77%) were achieved at the attenuation threshold of 4.7 mm^{-1} .²⁶ However, the ability of OCT to distinguish between precancerous lesions and cancer is still unclear, which is more urgent and challenging.

Texture features, as the objective information hidden in the images, are important identifications for distinguishing bio-tissues.^{27–31} In combination with machine learning algorithms, many studies have demonstrated that accurate recognition in OCT images can be effectively implemented. Based on texture features, Lingley-Papadopoulos et al.³² detected bladder cancer, and Adabi et al. applied several machine learning algorithms to distinguish human skin tissue in OCT images and demonstrated that computational models provided objective information to assist the clinicians in the diagnosis of abnormalities of cutaneous microstructure.³³ If a direct and effective diagnostic criterion can be used to differentiate OSCC from precancerous tissues, it will play a great role in assisting surgeons to determine the treatment. In this study, OCT images of oral normal mucosa, leukoplakia with epithelial hyperplasia (LEH), and OSCC obtained by our home-made swept-source OCT (SS-OCT) system were studied. The texture features-based artificial neural network (ANN) algorithm was used to distinguish different oral tissues, and the distinction performance was evaluated. The research provides the methodology for distinguishing OSCC and precancerous tissue from normal tissue, and it is proved that the appropriate algorithm can provide decision-making and improve diagnostic efficiency.

METHODS

SS-OCT system

The schematic and photograph of our home-made SS-OCT system are shown in Figure 1. It is based on Mach-Zehnder interferometry. The swept-source (HSL-2000-B; Santec) has a tunable spectral range of 87 nm and a sweep rate of 100 kHz. The output of the swept-source is about 20 mW. The galvanometer (GALVO) in the sample arm can realize the two-dimensional (2D) scan. The interference signal was detected by a balanced detector (Model 1817; New Focus) and then acquired by a data acquisition board (ATS-9351; Alazar Technologies Inc.). The axial resolution in air and lateral resolution are about 14 and 17 μm , respectively. The imaging speed reaches 100 frames per second, in which the image size of each frame is 1024×1000 . It takes about 10 s to collect a three-dimensional (3D) volumetric image consisting of 1000 2D images. The imaging range is $6 \times 6 \text{ mm}^2$. Details of the SS-OCT system can be found in our previous work.^{34,35}

Sample preparation

Fresh *ex vivo* tissues were scanned by SS-OCT system, which are from 19 patients undergoing oral-maxillofacial tumor resection at the Tianjin Stomatological Hospital.

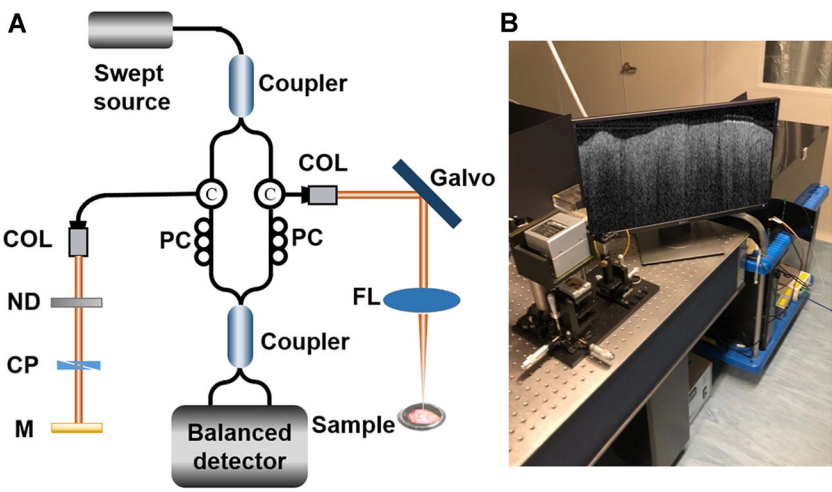


FIGURE 1 Schematic (A) and a photograph (B) of the home-made OCT system. C, circulator; COL, collimating lens; CP, compensation prisms; FL, focusing lens; ND, neutral density filter; PC, polarization controller; M, mirror

TABLE 1 The information of patients

	LEH	OSCC
Patients' number	5	14
Age (median [range])	62 [37–73]	60 [29–69]
Gender (male/female)	3/2	7/7

Abbreviations: LEH, leukoplakia with epithelial hyperplasia; OSCC, oral squamous cell carcinoma.

The study protocol was approved by the Ethics Committee of Tianjin Stomatological Hospital. The information of patients is given in Table 1. Among these patients, 14 are OSCC and 5 are LEH.

Taking into account the actual size of the excised tissue and the size of the lesion area, the OCT images of normal tissue regions were derived from the non-cancerous tissue (benign lesion) and the normal areas of some cancerous tissues (at least 1 cm away from cancerous margin) were carefully imaged. The tissue types of the scanned areas were finally confirmed after corresponding to the histopathological images, under the

guidance of an experienced surgeon (J.Z.). Based on this criterion, 14 normal areas were imaged. Tissues were placed in a sterile dish and immediately imaged after surgical resection to obtain the 2D and 3D OCT images.

The average sample size (from the top view) is about $3 \times 3 \text{ cm}^2$ and scans were performed several times in different locations. The diagnoses were ascertained by subsequent histopathology examination, and the final pathology reports were generated by a specialized pathologist (J.S.) at the Tianjin Stomatological Hospital.

As shown in Figure 2, the morphological structures of the OCT images of three kinds of oral tissues were identified and compared with the histopathological images. Figure 2A shows a typical OCT image of the normal oral mucosa, in which the epithelial layer (EP) and lamina propria (LP) are clearly distinguishable due to the differences in the intensity of optical scattering. The boundary between EP and LP, known as the basement membrane (BM), is indicated by a yellow dashed curve. The typical OCT image of LEH is shown in Figure 2B. The EP and LP are also clearly discernible, and the BM is intact, as indicated by a yellow dashed

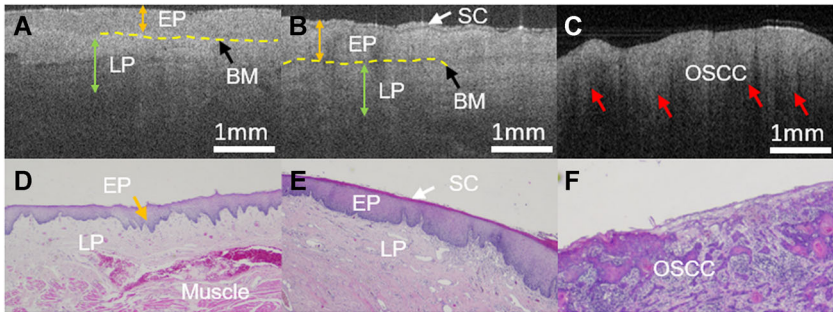


FIGURE 2 OCT images and the corresponding histopathological images of three oral tissues, including (A) and (D) normal mucosa, (B) and (E) LEH, and (C) and (F) OSCC. The yellow lines indicate the BM, which is the boundary between EP and LP. The red arrows show the nested and cord-like structures, which indicate the areas of OSCC. BM, basement membrane; EP, epithelial; LEH, leukoplakia with epithelial hyperplasia; LP, lamina propria; OCT, optical coherence tomography; OSCC, oral squamous cell carcinoma; SC, stratum corneum. The histology images were obtained by using an objective with the magnification of $\times 4$

curve. In addition, compared with the oral normal mucosa, the upper surface of dysplasia has the stratum corneum (SC) with strong optical scattering, as shown by the white arrow in Figure 2B. OSCC is an infiltrating carcinoma of the epithelium with varying degrees of squamous differentiation, consisting of irregular squamous epithelium with nests or cords. As shown in Figure 2C, the red arrows indicate the nested and cord-like structures. Moreover, the boundary between EP and LP is unclear, and the BM is destroyed. The corresponding histopathological images of normal mucosa, LEH, and OSCC are obtained by using an objective with the magnification of $\times 4$ and shown in Figure 2D–F, respectively.

Texture features extraction

According to our previous work,³⁶ appropriate texture features combination is beneficial to improve the performance of a training network. In our study, four sets of texture features are extracted from 2D OCT images. They are grey level co-occurrence matrix (GLCM), laws' texture measures (LTM), center symmetric auto-correlation (CSAC), and grey level histogram (GLH).

For GLCM,³⁷ 20 texture features, including the contrast, correlation, energy, homogeneity, and entropy in four different directions (0, 1), (−1, 1), (−1, 0), (−1, −1), are extracted.

For LTM,³⁸ five one dimensional convolution kernels are used by the outer product of these operators: Local average: $L5 = [1 \ 4 \ 6 \ 4 \ 1]$; Edge: $E5 = [-1 \ -2 \ 0 \ 2 \ 1]$; Spot: $S5 = [-1 \ 0 \ 2 \ 0 \ -1]$; Wave: $W5 = [-1 \ 2 \ 0 \ -2 \ 1]$; Ripple: $R5 = [1 \ -4 \ 6 \ -4 \ 1]$. Then, 20 2D convolution kernels were generated to extract texture features.

For CSAC,³⁹ 12 texture features are extracted by calculating the mean and standard deviation of grayscale texture covariance (SCOV), normalized SCOV, between-pair variance, within-pair variance, local variance, and variance ratio.

Six features belong to GLH, including mean, standard deviation, smoothness, third moment, uniformity, and entropy.⁴⁰

To optimize the combination of texture feature vectors, a principal component analysis (PCA) algorithm is employed to calculate the eigenvectors of the covariance matrix of the initial features and reduce feature dimensionality.⁴¹ The main idea of PCA is to map n -dimensional features to k -dimensional features ($k < n$), which are new orthogonal features, also known as principal components.

ANN establishment

An automatic computational model was designed in which different oral tissues are expected to be

classified by the BP-ANN. The ANN is a simplified computational model comprised of several layers of interconnected neurons. The construction of ANN is mainly considered the neural network layers, the number of nodes in each layer and the selection of training algorithms.^{42,43} Figure 3 shows the established ANN's architecture with an input layer, a hidden layer, and an output layer which are interconnected with different weights. The network was trained by the following parameters: learning rate, training epochs, convergence error, and minimum performance gradient. The values of these parameters are set to 0.001, 10000, 10^{-7} , and 10^{-7} , respectively. These parameters were obtained by trial-and-error.⁴³ Levenberg–Marquardt was selected as the optimization algorithm.

A total of 71 volumes for all tissue samples were obtained. Each OCT image contains only one type of tissue to be selected into the data set by comparing them with histopathologic images. A total of 3780 2D OCT images as a training set were used to train the back-propagation ANN (BP-ANN), including 1307 normal mucosa images, 757 LEH images, and 1716 OSCC images. The validation set of normal mucosa, LEH, and OSCC each included 150 OCT images. The remaining 946 OCT images (328 normal mucosa images, 194 LEH images, and 424 OSCC images) were used for the test to evaluate the performance of the algorithm.

The workflow of the sample acquisition and processing is shown in Figure 4. The algorithms of feature extraction and network training are executed on a desktop computer in our laboratory and the computer parameters include an eight-core Intel Xeon 3.5 GHz (E5-1620) processor and a 24 GB random-access memory. All algorithms in our study are run on MATLAB software.

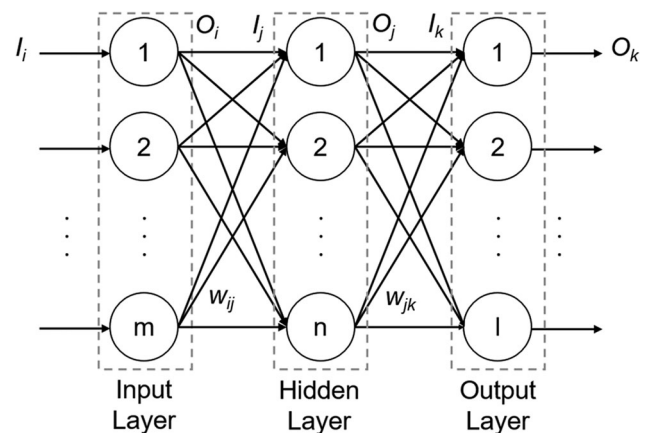


FIGURE 3 The typical ANN with one hidden layer. The number of nodes in hidden layer is 10 in our experiment. ANN, artificial neural network

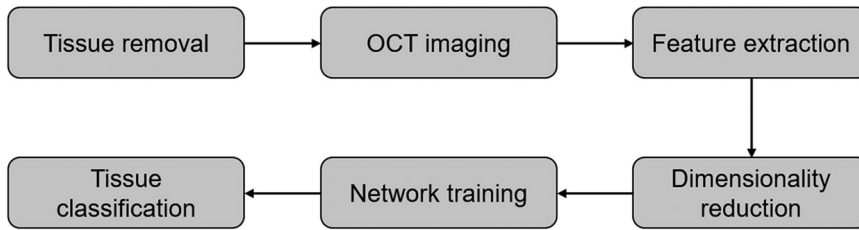


FIGURE 4 The workflow of the experiment. OCT, optical coherence tomography

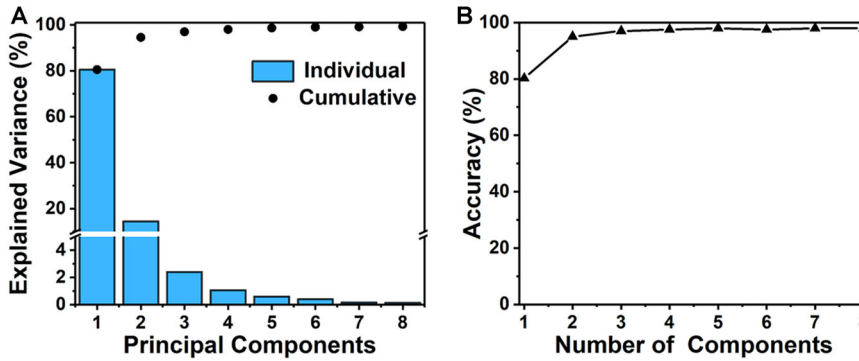


FIGURE 5 The quantitative analysis of texture features. (A) Explained variance of different principal components. (B) Accuracy distribution with the number of principal components

Quantitative analysis based on optical attenuation model

To compare with the texture-based classification method, a method based on optical attenuation model was also used to evaluate the classification performance of normal mucosa, LEH, and OSCC. According to Beer's law, a OCT signal in depth can be described in the following formula.^{26,44,45}

$$I(z) = A_1 \cdot \frac{\exp(-2\mu_t z)}{\left(\frac{z - z_{cf}}{z_r}\right)^2 + 1} + A_2. \quad (1)$$

Where $I(z)$ is the averaged signal profile versus depth z (in mm), z_{cf} is the position of the focal plane, μ_t is the attenuation coefficient, and z_r is the Rayleigh length. A_1 , A_2 , and μ_t are three fitting parameters.

RESULTS

Feature extraction

To reduce feature redundancy and improve computing speed, 58 texture features extracted from each whole OCT image in the data set were input to a PCA algorithm resulting in the first few principal components. To determine the number of principal components, the cumulative explained variance ratio was used to assess the ability of different principal components to contain valid information. Figure 5A shows the individual explained variance and the cumulative explained variance. It can be

found that most of the variance (80.59% of the variance) can be explained by the first principal component alone. The second principal component still holds 14.37% of the variance. The first eight principal components contain about 99% of the information. The average classification accuracies of the different number of principal components based on BP-ANN are shown in Figure 5B. With the increase in the number of principal components, the accuracy gradually increases and tends to be stable. Therefore, the first eight principal components were used in the following study.

Evaluation of ANN

After the PCA dimensionality reduction, BP-ANN-based classification and identification were implemented. The performance of this method was analyzed by a confusion matrix and the evaluation parameters, including sensitivity, specificity, precision, and accuracy were calculated, which are shown in Table 2. The sensitivity and specificity of normal mucosa, LEH, and OSCC, are 98.17% / 98.38%, 93.81% / 98.54%, and 98.11% / 99.04%, respectively, their precisions are 96.99%, 94.30%, and 98.81%, respectively, and their accuracies are 98.31%, 97.57%, and 98.63%, respectively. Their average classification accuracy based on the BP-ANN is 98.17%. The average time of training is 283.15 s and the identification time is 60.34 μ s.

As shown in Figure 6, the receiver operating characteristic (ROC) curves that indicate the relationship between the true-positive and false-positive rates were further depicted to assess the classification performance

TABLE 2 Classification results of texture-based BP-ANN for the test set

	Actual classes		
	Normal (328)	LEH (194)	OSCC (424)
Predicted classes			
Normal (332)	322	8	2
LEH (193)	5	182	6
OSCC (421)	1	4	416
Evaluation parameters			
Sensitivity (%)	98.17	93.81	98.11
Specificity (%)	98.38	98.54	99.04
Precision (%)	96.99	94.30	98.81
Accuracy (%)	98.31	97.57	98.63

Abbreviations: LEH, leukoplakia with epithelial hyperplasia; OSCC, oral squamous cell carcinoma.

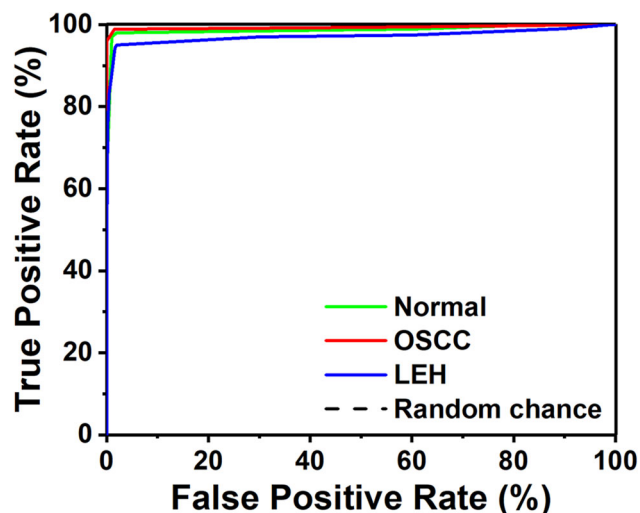


FIGURE 6 The ROC curves for the classification of different oral tissues. The green curve is the result of normal mucosa, the red curve is the result of OSCC, the blue curve is the result of LEH, and the dashed line is the random chance. The AUCs for differentiating normal mucosa, OSCC, and LEH is 0.98, 0.99, and 0.96, respectively. LEH, leukoplakia with epithelial hyperplasia; OSCC, oral squamous cell carcinoma; ROC, receiver operating characteristic curve

of these three oral tissues. The area under the curve (AUC) is a quantitative performance measure, with an AUC of 1.0 corresponding to perfect results and 0.5 indicating equal to random performance. It can be found that all curves are very close to the top left corner, and the AUCs for normal mucosa, OSCC, and LEH are 0.98, 0.99, and 0.96, respectively, which proves the texture features-based BP-ANN method is excellent for distinguishing OSCC, precancerous tissue, and normal mucosa.

Optical attenuation model

Figure 7A shows the OACs distribution of three kinds of oral tissues. It can be seen that normal mucosa (green) and OSCC (red) can be distinguished effectively, and normal mucosa and LEH (blue) can also be distinguished, whereas LEH and OSCC cannot be distinguished because of too much overlap. Further, ROC curves were drawn in Figure 7B and AUCs were calculated to quantitatively evaluate classification performance. The AUCs for normal mucosa, OSCC, and LEH are 0.91, 0.95, and 0.78, respectively.

DISCUSSIONS

It is demonstrated from the high accuracies, sensitivities, and specificities that texture features can be used as typical indicators of normal mucosa, LEH, and OSCC in OCT images, and BP-ANN is a convincing tool to be used in differentiating oral precancerous and cancerous tissues.

For biological tissues without obvious structural layers, such as OSCC, the texture feature-based methods can realize effective identification. In recent years, texture-based methods have been introduced widely. Furthermore, to avoid feature redundancy, the selection of texture features is critical. According to the previous work,³⁶ different combinations of texture features have different classification effects. In this study, four sets of texture features were used and the superior results were obtained with an average accuracy of 98.17%. By contrast, the classification accuracy of single texture feature was calculated, and the average classification accuracies are 85.25%, 77.63%, 80.25%, and 72.54% for GLCM, LTM, CSAC, and GLH, respectively.

Compared with the texture-based classification method, the optical attenuation model achieves a lower AUCs than the method in this study, which indicates the optical attenuation model has limitations in the multi-classification task. Therefore, it is necessary to add quantitative indicators or introduce machine learning algorithms for the multi-classification questions.

The validity of the data set for ANN training is worth discussing. According to the imaging characteristics of OCT system, there are differences between adjacent OCT images, even if it comes from the same patient. To further avoid similarities, 2D OCT images with nonadjacent B-scans were selected. Therefore, each image is input into the network as a separate sample. It may exist some limitations for mixing patient data between the testing and validation set. In this case, the test results about the evaluation parameters may be slightly higher than those without mixing the testing and validation set. In addition, to verify the generalization ability of our model, we tested the OSCC data of another patient who was not employed in this study. The classification accuracy is

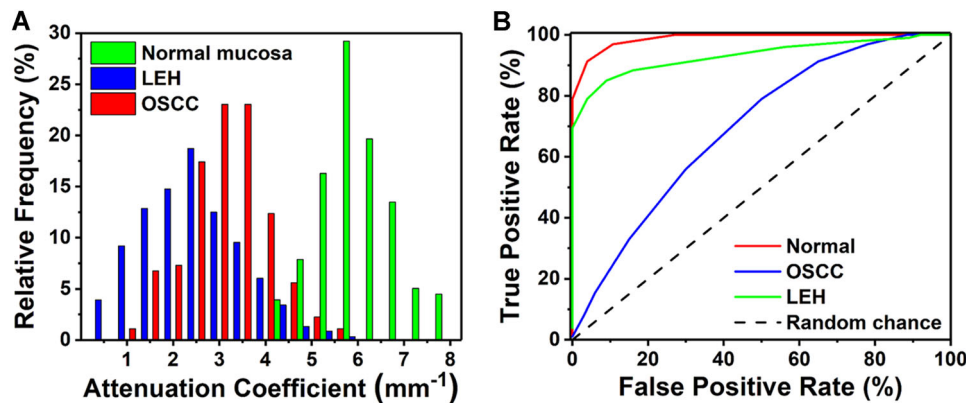


FIGURE 7 Classification results of the three kinds of oral tissues based on optical attenuation model. (A) is their frequency distribution of OACs. (B) is the ROC curves of OACs for different sample combinations. The AUCs for differentiating normal mucosa, OSCC, and LEH are 0.91, 0.95, and 0.78, respectively. LEH, leukoplakia with epithelial hyperplasia; OAC, optical attenuation coefficient; OSCC, oral squamous cell carcinoma; ROC, receiver operating characteristic curve

97.33% (292/300 images), which can prove the robustness of the model to a certain extent.

Considering the complexity of the oral environment, noninvasive, high resolution, and real-time detection technology will provide necessary conditions for the diagnosis of oral diseases. Compared with other imaging methods, OCT has advantages that make it more suitable for oral mucosal tissue imaging. *In vivo* early screening is also possible by using portable oral probes.³⁵ In addition, with the acceleration of real-time OCT acquisition and the increasing number of data samples, it is very important for the direct analysis of OCT images. On this basis, an appropriate auxiliary diagnostic approach will improve the clinical outcome and the patients' prognosis recovery. By comparing with the optical attenuation model-based method, the texture-based method has superior performance and more extensive applicability.

Our ultimate goal is to provide a reliable, affordable, and accurate OCT-based tool for the diagnosis of human oral diseases, especially oral cancer. In the future, on the one hand, we will develop a portable probe with higher resolution and field of view, as well as an OCT system for validation in a large number of human patients, enabling the clinical translation from bench to bedside. On the other hand, we will further study the recognition algorithms of OCT images in the diagnosis of oral diseases by combining deep learning. It is possible to improve the screening and diagnosis of oral cancers.

CONCLUSION

In this study, fresh *ex vivo* oral tissues were imaged intraoperatively by a home-made SS-OCT system. Then, a texture-based identification and classification method was studied to differentiate oral normal mucosa, pre-cancerous and cancerous tissues. The diagnostic

performance was evaluated and it was found that the texture-based method had higher diagnostic utility and wider applicability, compared with the optical attenuation model. Results proved that an appropriate algorithm can help to improve the screening and diagnosis of oral cancers.

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CONFLICT OF INTERESTS

The authors declare that there are no conflict of interests.

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